

OOSTENDE PIER, Belgium

WMO: 064080

Lat: 51.239N

Lon: 2.917E

Elev: 15

StdP: 101.14

Time Zone: 1.00 (EUC)

Period: 90-05

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-4.6	-2.5	-7.6	2.0	-3.8	-5.5	2.4	-1.2	20.7	9.6	19.0	10.2	3.7	70	0.706

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.9	25.7	20.4	23.6	19.3	22.2	18.5	20.9	24.3	19.9	22.8	19.0	21.9	4.0	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.7	14.5	22.5	18.7	13.5	21.8	17.8	12.8	21.0	60.4	24.5	56.8	23.0	54.0	21.9	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.2	15.5	13.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			-6.3	23.0	3.3	2.1	-8.7	24.5	-10.6	25.7	-12.4	26.9	-14.8	28.5	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.3	5.3	6.1	7.7	9.5	13.0	15.5	17.9	18.7	15.5	12.5	8.4	5.0		
	DBStd	5.52	4.00	2.97	2.24	2.58	2.94	2.60	2.19	2.43	2.04	3.01	2.93	3.51		
	HDD10.0	612	148	111	77	39	5	0	0	0	0	13	62	157		
	HDD18.3	2644	406	341	331	266	168	93	34	23	89	181	298	414		
	CDD10.0	1082	1	2	4	23	98	165	246	271	165	90	14	2		
	CDD18.3	73	0	0	0	0	2	8	23	36	4	0	0	0		
	CDH23.3	201	0	0	0	0	14	26	50	106	6	0	0	0		
	CDH26.7	45	0	0	0	0	4	6	7	28	0	0	0	0		
(14) Wind	WSAvg	6.0	6.6	7.3	6.2	5.9	5.8	5.8	5.6	5.4	5.0	5.6	6.4	6.6		
(15) Precipitation	PrecAvg	777	62	55	45	43	56	66	70	74	66	75	87	77		
	PrecMax	962	120	97	93	78	143	148	160	158	168	150	174	182		
	PrecMin	570	3	14	9	5	12	24	15	7	5	19	10	14		
	PrecStd	106	28	25	24	21	28	31	34	45	43	32	36	42		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.6	14.5	16.8	20.7	24.3	26.8	27.8	30.5	23.9	21.2	16.1	13.0		
		MCWB	11.1	11.0	10.9	13.2	17.3	18.1	20.1	21.5	18.2	16.5	13.1	11.9		
	2%	DB	11.5	12.1	13.7	17.0	20.7	23.3	24.9	26.1	21.4	18.6	14.0	11.9		
		MCWB	10.1	9.8	10.0	12.1	16.1	17.6	20.2	21.6	17.4	15.3	12.4	10.8		
	5%	DB	10.7	10.9	11.9	14.4	18.6	20.8	22.9	23.9	19.9	17.3	12.9	10.6		
		MCWB	9.3	8.9	9.3	11.2	15.3	16.9	18.9	20.3	16.6	15.0	11.3	9.4		
	10%	DB	9.8	9.8	10.6	12.8	17.1	19.2	21.6	22.5	18.6	16.3	12.0	9.5		
		MCWB	8.5	8.1	8.6	10.5	14.3	16.1	18.3	19.3	15.6	14.2	10.5	8.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.2	11.7	11.7	14.7	19.1	19.5	21.6	24.2	19.2	17.2	13.7	12.0		
		MCDB	12.2	13.8	15.0	17.8	22.5	25.1	25.3	24.9	22.2	19.4	14.7	12.8		
	2%	WB	10.3	10.4	10.6	12.9	17.0	18.2	20.3	21.6	18.1	16.1	12.6	10.8		
		MCDB	11.3	11.8	12.7	16.2	18.8	22.3	23.4	25.2	20.5	18.1	13.6	11.8		
	5%	WB	9.5	9.1	9.7	11.7	15.5	17.3	19.3	20.5	17.1	15.3	11.6	9.7		
		MCDB	10.5	10.4	11.5	14.0	17.8	20.3	22.2	23.7	19.4	17.0	12.6	10.5		
	10%	WB	8.6	8.3	9.0	10.7	14.4	16.3	18.4	19.6	16.1	14.5	10.8	8.5		
		MCDB	9.6	9.6	10.5	12.5	16.9	18.9	21.2	22.3	18.1	16.1	11.9	9.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	3.6	3.7	3.8	4.3	4.7	4.3	4.6	4.9	4.7	4.5	3.6	3.5		
		MCDBR	4.5	5.1	6.7	7.6	7.5	7.4	7.1	7.9	7.2	5.9	4.3	4.2		
	5% WB	MCWBR	4.1	4.2	4.5	4.9	5.0	4.4	4.5	4.5	4.3	3.8	3.8	3.8		
		MCWBR	4.5	4.4	5.2	6.3	7.0	6.5	6.4	7.3	6.0	5.2	4.0	4.0		
(40) Clear-Sky Solar Irradiance	taub		0.334	0.353	0.390	0.413	0.423	0.420	0.422	0.413	0.391	0.370	0.345	0.329		
		taud	2.370	2.357	2.257	2.226	2.251	2.280	2.284	2.326	2.373	2.407	2.381	2.348		
	Ebn at Noon		674	765	798	824	833	838	829	816	792	735	651	612		
		Edh at Noon	64	83	110	127	129	127	124	114	97	79	63	56		
(44) All-Sky Solar Radiation	RadAvg		0.77	1.47	2.70	4.41	5.15	5.58	5.28	4.50	3.32	1.90	0.90	0.58		
	RadStd		0.07	0.27	0.34	0.40	0.38	0.44	0.53	0.38	0.22	0.16	0.09	0.07		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (72 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page